

sword_doctor.sh — VPN-LAN sword bridge preflight doctor

Revision: 1 **Last modified:** 2026-07-01T16:00:00Z **Companion to:** [scripts/sword_doctor.sh](#) **Authority:** Inherits constitution/Constitution.md per §11.4.35 · §11.4.18 script-documentation mandate.

Overview

sword_doctor.sh is the Phase-0 preflight for the VPN-LAN service-access feature (design: [docs/design/vpn_lan_access/PLAN.md](#) §3 + §5). It resolves the env-var **bridge contract** that decouples helix_proxy from the sibling sword_toolkit VPN bridge (§11.4.28) and emits a single machine-parseable verdict telling every downstream VPN-LAN test whether the bridge is **UP**, **DOWN** (operator-blocked), or **MISCONFIGURED**.

The doctor never fakes an UP verdict (§11.4.6): a BRIDGE: UP requires the operator-supplied health probe to exit 0 **and** the remote host to answer a smoke probe. When the bridge is down or unreachable, downstream tests must honestly SKIP with network_unreachable_external (§11.4.3 / §11.4.68 / §11.4.69) — a fake PASS is forbidden.

Prerequisites

- POSIX sh (validated under sh -n and bash -n, §11.4.67).
- [tests/lib/sword_bridge.sh](#) — the contract library the doctor sources.
- ping (preferred) or nc for the remote-host smoke probe.
- The 6 bridge-contract environment variables set (PLAN.md §3). Real values live in a **gitignored** .env; the tracked [.env.example](#) documents the shape (names + illustrative paths only — no secrets, §11.4.10).

Env var	Meaning
HELIX_SWORD_DIR	Path to the sibling bridge project

Env var	Meaning
HELIX_BRIDGE_CONNECT	Command that brings the VPN up
HELIX_BRIDGE_DISCONNECT	Command that tears the VPN down
HELIX_BRIDGE_HEALTH	Health probe (exit 0 == up) — authoritative signal
HELIX_BRIDGE_SUBNET	Reachable remote subnet (CIDR)
HELIX_BRIDGE_HOST	Known remote host for smoke reachability

Usage

```
# 1. Point the bridge at your sword_toolkit checkout (once):
cp .env.example .env          # .env is gitignored – never commit
                                it
$EDITOR .env                  # fill in real HELIX_* values

# 2. Source .env into the environment and run the doctor:
set -a; . ./env; set +a
scripts/sword_doctor.sh; echo "exit=$?"
```

Optional override (used by tests): SVORD_BRIDGE_LIB=/path/to/sword_bridge.sh.

Verdicts + exit codes

The final line always matches ^BRIDGE: ; parse it, ignore the sword-doctor: diagnostic lines above it.

Verdict line	Exit	Meaning
BRIDGE: UP	0	Contract resolved, hooks executable, health exit 0, host reachable
BRIDGE: SKIP:network_unreachable_external	2	Bridge down / host unreachable — OPERATOR-BLOCKED (\$11.4.68)
BRIDGE: SKIP:host_probe_unavailable	2	No ping/nc present — reachability cannot be confirmed

Verdict line	Exit	Meaning
		(honest, not UP)
BRIDGE: MISCONFIGURED:env_unset	3	One or more contract vars unset/empty
BRIDGE: MISCONFIGURED:hook_not_executable:<vars>	3	A hook's first-token path is missing / not executable
BRIDGE: MISCONFIGURED:bridge_lib_missing	3	svord_bridge.sh not found

Downstream test harness convention: exit 0 → run the live path; exit 2 → SKIP with `network_unreachable_external`; exit 3 → SKIP/FAIL as a configuration error (fix the contract first).

Edge cases

- **Contract unset** → MISCONFIGURED:env_unset (exit 3), NOT UP. The doctor lists exactly which vars are unset on the `svord-doctor`: diagnostic lines.
- **Hook path not executable** → MISCONFIGURED:hook_not_executable:<vars> (exit 3). Only the first whitespace token of each hook command is checked, so hooks may carry arguments (e.g. `.../svord-ssh-health --quiet`).
- **Unexpanded \${VAR} in a hook value** → treated as a literal path; if it does not resolve to an existing file the doctor reports misconfigured. Source your `.env` with `set -a` so the shell expands the values first. The doctor never evals hook contents (injection-safe, §11.4.10).
- **Health probe exits non-zero** → SKIP:network_unreachable_external (exit 2). This is the authoritative down signal; the host probe is not reached.
- **Host unreachable but health up** → SKIP:network_unreachable_external (exit 2). ICMP-filtered hosts fall back to a TCP probe (`nc`) before being declared down.
- **Neither ping nor nc present** → SKIP:host_probe_unavailable (exit 2) — the doctor refuses to claim UP it cannot confirm (§11.4.6).

Internal behaviour

1. Resolve + source `tests/lib/svord_bridge.sh` (SVORD_BRIDGE_LIB override).
2. `bridge_load` — every one of the 6 contract vars set + non-empty, else MISCONFIGURED:env_unset (3).

3. For `HELIX_BRIDGE_CONNECT` / `_DISCONNECT` / `_HEALTH`: first token must be an existing, executable file, else `MISCONFIGURED:hook_not_executable` (3).
4. `bridge_up` — run `HELIX_BRIDGE_HEALTH`; non-zero → `SKIP:network_unreachable_external` (2).
5. Smoke-probe `HELIX_BRIDGE_HOST` (ping -c1 -w1, TCP fallback) → `unreachable SKIP:network_unreachable_external` (2); no probe tool `SKIP:host_probe_unavailable` (2).
6. Otherwise `BRIDGE`: `UP` (0).

The doctor is **invocation-only**: it runs the operator's hooks and one smoke probe, writes nothing, and modifies neither the bridge project nor any remote host (§11.4.122).

Related scripts

- [tests/lib/sword_bridge.sh](#) — the sourceable contract library (`bridge_load`, `bridge_up`, `bridge_require`, `bridge_subnet`, `bridge_host`).
- [.env.example](#) — the tracked contract template (§3).
- [docs/design/vpn_lan_access/PLAN.md](#) — the feature design (§3 contract, §5 Phase 0).

Anti-bluff evidence

Phase-0 discrimination proof (same local stub bridge, only the health exit code differs → verdict flips `UP` ↔ `SKIP`) captured under `qa-results/vpn_lan/phase0/<UTC-ts>/`:

Condition	Verdict	Exit
contract UNSET	<code>MISCONFIGURED:env_unset</code>	3
stub health exit 0, host 127.0.0.1	<code>UP</code>	0
stub health exit 1 (RED teeth)	<code>SKIP:network_unreachable_external</code>	2

Last verified date

2026-07-01 — `sh -n + bash -n PASS`; 3-verdict discrimination proof + lib primitive proof captured under `qa-results/vpn_lan/phase0/20260701T155245Z/`.